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Roll No. .....

SIXTH SEMESTER

B.E. (COE)

MID SEMESTER EXAMINATION ***MARCH*** **2010**

COE-315 ADVANCE COMPUTER ARCHITECTURE

Time: 1 Hour 30 Minutes ***Max. Marks : 20***

Note : Answer ***ALL*** questions.
Assume suitable missing data, if any.

1[a] Define parallel computers and distributed computers. What are their purpose?

[b] What are the limits of Network model as a parallel machine?

[c] Give a CREW-PRAM algorithm to compute Pascal triangle :

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots\ldots\ldots, \binom{n}{n}$$

for n =0, 1, 2, 3, 4, 5, 6

2+2+2

2 For the following reservation table of a pipeline processor, give the list of forbidden latencies, the collision vector, and obtain MAL. Have you obtained the optimal MAL? If not, can you achieve it by delay insertion? Justify your answer.

Stage \ Time	1	2	3	4	5	6	7
S1	X			X			
S2		X					X
S3			X				
S4					X		
S5						X	

OR

2[a] What do you understand by
(i) UMA-Uniform memory access architecture
(ii)Distributed shared-memory architecture

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[b] Find the diameter, Bisection width, node degree and design simple routing algorithm for each of the following INs:

(i) Hypercube (n-cube)

(ii) 2-D mesh of n-node

2+4=6

3[a] 95% of a program's execution time occurs inside a loop that can be executed in parallel. What is the maximum speedup we should expect from a parallel version of the program executing on 8 CPUs?

[b] An application running on 10 processors spends 3% of its time in serial code. What is the scaled speedup of the application?

2+2=4

4 Write the parallel search algorithm for tree. Illustrate it for the list L = {24, 14, 35, 27, 17, 19, 26, 19} and x = 19. Calculate the time taken by your algorithm.

OR

4 Write the parallel algorithm for matrix multiplication on a cube-connected SIMD computer with n^3 processor, assume that matrices are of order n×n. Also, obtain the time taken by your algorithm. **4**

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